

## Introduction

Subgroup meta-analysis partitions the included studies according to a pre-specified categorical moderator — intervention intensity (low vs high dose), population (paediatric vs adult), risk of bias (low vs unclear vs high), or any clinically meaningful split — and pools within each level. The within-subgroup pooled effects describe how the intervention behaves in each stratum, and a formal between-subgroup test assesses whether the moderator explains a measurable share of between-study heterogeneity. Subgroup analysis is the standard tool for categorical effect modification in evidence synthesis and is required by the Cochrane Handbook whenever the protocol pre-specifies a moderator hypothesis.

## Prerequisites

A working understanding of pooled meta-analysis (preferably random-effects), the heterogeneity statistics  $\tau^2$ ,  $I^2$ , and  $Q$ , and the difference between within-group and between-group sources of variation in stratified analyses.

## Theory

Within each subgroup  $g = 1, \dots, G$  the studies are pooled to obtain  $\hat{\theta}_g$  and its standard error. The between-subgroup heterogeneity statistic is

$$Q_B = \sum_g w_g (\hat{\theta}_g - \hat{\theta})^2,$$

with  $G - 1$  degrees of freedom; a significant  $Q_B$  indicates the moderator accounts for some of the between-study variance. Equivalently, fitting a mixed-effects meta-regression with subgroup as a categorical moderator produces a  $Q_M$  test that recovers exactly this comparison and is the modern preferred implementation, because it can also share a common  $\tau^2$  across subgroups for more stable inference.

## Assumptions

The moderator is pre-specified rather than chosen post hoc, within-subgroup studies are reasonably homogeneous (small within-group  $\tau^2$ ), and each subgroup contains enough studies — typically at least four to five — to support a stable within-group pool. Subgroup analyses with only one or two studies per stratum are largely uninformative and should be reported only descriptively.

## R Implementation

```
library(metafor)

set.seed(2026)
k <- 16
subgroup <- factor(rep(c("low_dose", "high_dose"), each = k/2))
yi <- c(rnorm(k/2, 0.2, 0.15),
        rnorm(k/2, 0.5, 0.15))
vi <- 1 / sample(50:200, k, replace = TRUE)

df <- data.frame(yi, vi, subgroup)

lapply(split(df, df$subgroup), function(d) {
  rma(yi, vi, data = d, method = "REML")$b
})
```

```
res_mod <- rma(yi, vi, mods = ~ subgroup, data = df, method = "REML")
summary(res_mod)
```

## Output & Results

The script returns per-subgroup pooled estimates from independent random-effects pools and the joint mixed-effects meta-regression with subgroup as moderator. The latter reports the omnibus  $Q_M$  test for between-subgroup heterogeneity, the per-level coefficients, and the residual heterogeneity remaining after the moderator is included — the canonical reporting set for subgroup analyses.

## Interpretation

A reporting sentence: “Pre-specified subgroup analysis showed significant between-subgroup heterogeneity ( $Q_M = 18.4$ ,  $df = 1$ ,  $p < 0.001$ ): high-dose studies had a pooled standardised mean difference of 0.52 (95 % CI 0.39 to 0.65) compared with 0.18 (0.08 to 0.28) in low-dose studies, supporting the hypothesis that dose moderates the treatment effect. The moderator explained 47 % of the between-study heterogeneity ( $R^2$ ); residual  $I^2$  was 18 %.” Always report both the within-subgroup pools and the between-subgroup test.

## Practical Tips

- Pre-specify all subgroup analyses in the PROSPERO protocol; post-hoc subgrouping is exploratory and should be flagged as such — never as primary inference.
- Small per-subgroup study counts produce unstable pooled estimates; the Cochrane Handbook recommends at least 10 studies per subgroup before drawing conclusions, and warns explicitly against single-stratum subgroup pools.
- Prefer meta-regression over subgroup analysis whenever the moderator is continuous (age, dose, year); dichotomising loses information and depends on an arbitrary cutpoint.
- Report the between-subgroup  $Q_M$  or  $Q_B$   $p$ -value as the primary inference; comparing within-subgroup  $p$ -values across strata (“significant in one, not the other”) is a well-known statistical fallacy because tests in small subgroups are underpowered.
- Use a shared  $\tau^2$  across subgroups (the default in mixed-effects meta-regression) rather than separate estimates per subgroup unless heterogeneity is clearly different in each; the shared specification is more efficient and more conservative.
- When subgroup analysis is exploratory, apply a multiplicity adjustment or interpret with explicit caution; the false-positive rate inflates rapidly with the number of subgroup tests.

## R Packages Used

`metafor::rma()` with `mods = ~ factor(subgroup)` for the canonical mixed-effects implementation; `meta::metabin()` and `meta::metacont()` with `subgroup =` for built-in subgroup pooling and forest-plot grouping; `dmatar` for Knapp–Hartung-adjusted subgroup analyses; `robumeta` for robust-variance subgroup pooling under correlated effects.

## For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

### See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis